

Mathematical Induction

Notes By Pr. El Hadiq Zouhair

13.1 The well-ordering principle

Theorem 13.1 (Well-ordering principle, WOP). Every nonempty subset of $\mathbb{N}$ has a least element.

This innocuous statement is the bedrock: it is logically equivalent to the principle of induction, and powers “minimal counterexample” arguments — if a statement failed for some n , there would be a *smallest* failing n , which one then contradicts.

13.2 Weak induction

Theorem 13.2 (Principle of mathematical induction, PWI). Let $P(n)$ be a predicate on $\mathbb{N}$. If

- (i) **Base case:** $P(0)$ holds (or $P(n_0)$ for a starting point n_0), and
 - (ii) **Inductive step:** for all $n \geq n_0$, $P(n) \Rightarrow P(n+1)$,
- then $P(n)$ holds for all $n \geq n_0$.

WOP implies PWI. Suppose (i) and (ii) hold but P fails somewhere. By WOP let m be the least failure. $m \neq n_0$ by (i), so $m-1 \geq n_0$ and $P(m-1)$ holds (minimality). By (ii), $P(m)$ holds — contradiction. (Conversely PWI implies WOP: see Exercise 13.10.) $\square$

The template. To prove $\forall n \geq n_0, P(n)$: (1) *state* $P(n)$ clearly; (2) verify $P(n_0)$; (3) fix $n \geq n_0$, assume the *inductive hypothesis* $P(n)$, and derive $P(n+1)$ — pointing precisely at where the hypothesis is used; (4) conclude by induction.

13.3 Summation identities

Example 13.3. $\sum_{i=1}^n i = \frac{n(n+1)}{2}$ for $n \geq 1$. *Base:* $1 = \frac{1 \cdot 2}{2}$. *Step:* assuming the identity for n ,

$$\sum_{i=1}^{n+1} i = \frac{n(n+1)}{2} + (n+1) = (n+1) \left(\frac{n}{2} + 1 \right) = \frac{(n+1)(n+2)}{2}. \quad \square$$

Example 13.4. $\sum_{k=0}^n 2^k = 2^{n+1} - 1$: base $1 = 2 - 1$; step: add 2^{n+1} to both sides: $2^{n+1} - 1 + 2^{n+1} = 2^{n+2} - 1$. Similarly $\sum_{k=1}^n k \cdot k! = (n+1)! - 1$: step uses $(n+1)!(n+2) - 1 - \dots$ — see Exercise 13.2. The sums $\sum k^2 = \frac{n(n+1)(2n+1)}{6}$ and $\sum k^3 = \left(\frac{n(n+1)}{2}\right)^2$ follow the identical pattern (Chapter 9).

13.4 Inequalities

Example 13.5. $2^n > n^2$ for all $n \geq 5$. *Base:* $2^5 = 32 > 25$. *Step:* for $n \geq 5$, assume $2^n > n^2$. Then

$$2^{n+1} = 2 \cdot 2^n > 2n^2 = n^2 + n \cdot n \geq n^2 + 5n > n^2 + 2n + 1 = (n+1)^2,$$

using $n \geq 5 \Rightarrow 5n > 2n + 1$. Note the base case 5 is essential: the claim is false for $n = 2, 3, 4$.

Example 13.6. $n! \geq 2^n$ for $n \geq 4$: base $24 \geq 16$; step: $(n+1)! = (n+1) \cdot n! \geq (n+1)2^n \geq 2 \cdot 2^n = 2^{n+1}$ since $n+1 \geq 2$. Two classical companions provable the same way: Bernoulli's inequality $(1+x)^n \geq 1+nx$ for $x \geq -1$, and $\ln(n+1) \leq H_n = \sum_{i=1}^n \frac{1}{i}$.

13.5 Divisibility

Example 13.7. $3 \mid (n^3 + 2n)$ for all $n \geq 0$. *Base:* $3 \mid 0$. *Step:*

$$(n+1)^3 + 2(n+1) = n^3 + 3n^2 + 3n + 1 + 2n + 2 = (n^3 + 2n) + 3(n^2 + n + 1),$$

a sum of two multiples of 3 (the first by hypothesis).

13.6 Sets and geometry

Example 13.8 (Subsets). A set with n elements has 2^n subsets. *Base:* $\mathcal{P}(\emptyset) = \{\emptyset\}$, of size $1 = 2^0$. *Step:* let $|A| = n+1$, pick $x \in A$, $A' = A \setminus \{x\}$. Subsets of A split into subsets of A' and those plus x : $2^n + 2^n = 2^{n+1}$ by hypothesis.

Example 13.9 (Triangulation). Every convex polygon with $n \geq 3$ vertices can be divided into $n-2$ triangles by non-crossing diagonals. *Base:* a triangle, $1 = 3-2$. *Step:* in a convex

$(n+1)$ -gon, the diagonal from v_1 to v_n cuts off triangle $v_1v_nv_{n+1}$ and leaves a convex n -gon, triangulated by hypothesis into $n-2$ triangles: total $n-1 = (n+1)-2$.

13.7 Strong induction

Theorem 13.10 (Strong induction). If $P(n_0)$ holds and, for all $n \geq n_0$,

$$(P(n_0) \wedge P(n_0+1) \wedge \cdots \wedge P(n)) \Rightarrow P(n+1),$$

then $P(n)$ holds for all $n \geq n_0$. Strong and weak induction are equivalent in power, but strong induction is the natural tool when $P(n+1)$ depends on values *further back* than n .

Example 13.11 (Prime factorisation). Every integer $n \geq 2$ is a product of primes. If $n+1$ is prime, done. Otherwise $n+1 = ab$ with $2 \leq a, b \leq n$; by the strong hypothesis both a and b are products of primes, hence so is $n+1$. (Weak induction is helpless here: a, b are far below $n+1$.)

Example 13.12 (Fibonacci bound). $F_n < 2^n$ for all $n \geq 0$: bases $F_0 = 0 < 1$, $F_1 = 1 < 2$; step ($n \geq 1$): $F_{n+1} = F_n + F_{n-1} < 2^n + 2^{n-1} < 2^{n+1}$, using the hypothesis at *two* previous points.

Example 13.13 (Postage stamps). Every amount $n \geq 12$ can be paid with 4- and 5-dirham stamps. Bases: $12 = 4 + 4 + 4$, $13 = 4 + 4 + 5$, $14 = 4 + 5 + 5$, $15 = 5 + 5 + 5$. Step for $n+1 \geq 16$: by strong hypothesis $(n+1) - 4 \geq 12$ is payable; add one 4-stamp.

13.8 Structural induction

For recursively defined structures (Chapter 14) — formulas, strings, trees — induction follows the structure: prove the property for the base objects, and show each construction rule preserves it. Example: every wff of propositional logic has as many left as right parentheses — true for atoms (none), and each formation rule adds one of each.

13.9 Pitfalls

Example 13.14 (All horses are the same colour). “Proof” that any n horses share a colour: for the step, remove one horse — the remaining n share a colour by hypothesis; remove a different one — same conclusion; the overlap forces all $n+1$ to match. The argument *fails* at $n+1 = 2$: the two groups of one horse do not overlap, so nothing links them. One broken link ($P(1) \rightarrow P(2)$) destroys the whole chain.

Always check: (1) the base case is actually verified (a missing or false base “proves” $\sum_{i=1}^n i = \frac{n(n+1)}{2} + 1$ too, since the step goes through!); (2) the inductive step is valid for every n from the base upward, including the very first; (3) the inductive hypothesis is genuinely used — otherwise the proof is direct, or wrong.

13.10 Induction and programs

Induction is the mathematics of loops and recursion: proving that a recursive program is correct *is* an induction on the input (see Chapter 14), and a loop is proved correct via a *loop invariant* — a property maintained by every iteration, i.e. an inductive step. Example: in the iterative factorial, the invariant “after k iterations the accumulator equals $k!$ ” is established at $k = 0$ and preserved each pass, so on exit (at $k = n$) the result is $n!$.

13.11 Summary

WOP, weak induction and strong induction are equivalent foundations. The template — state $P(n)$, verify the base, derive $P(n + 1)$ using the hypothesis visibly — proves summation identities, inequalities (mind the true base case), divisibility facts, set-theoretic and geometric claims. Strong induction handles recurrences reaching far back (prime factorisation, Fibonacci); structural induction handles recursively built objects; and loop invariants are induction wearing a programmer’s coat.

Exercises

Exercise 13.1 EASY

Prove by induction: $\sum_{i=1}^n (2i - 1) = n^2$ for $n \geq 1$.

Exercise 13.2 EASY

Prove by induction: $\sum_{k=1}^n k \cdot k! = (n + 1)! - 1$.

Exercise 13.3 EASY

Prove by induction: $\sum_{k=1}^n \frac{1}{k(k+1)} = \frac{n}{n+1}$.

Exercise 13.4 MEDIUM

Prove that $n! > 3^n$ for all $n \geq 7$.

Exercise 13.5 MEDIUM

Prove that $6 \mid (n^3 - n)$ for all $n \geq 0$.

Exercise 13.6 MEDIUM

Prove by induction that $\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2}\right)^2$.

Exercise 13.7 MEDIUM

Prove Bernoulli’s inequality: $(1 + x)^n \geq 1 + nx$ for all real $x \geq -1$ and $n \in \mathbb{N}$. Where does the step use $x \geq -1$?

Exercise 13.8 MEDIUM

A chocolate bar is an $m \times n$ grid of squares. Prove (strong induction on the number of squares) that exactly $mn - 1$ breaks along grid lines are needed to reduce it to unit squares.

Exercise 13.9 HARD

Prove by strong induction that every $n \geq 1$ has a binary representation: it can be written as $\sum_i b_i 2^i$ with $b_i \in \{0, 1\}$.

Exercise 13.10 HARD

Deduce the well-ordering principle from weak induction: if $S \subseteq \mathbb{N}$ has no least element, prove by strong induction that $S = \emptyset$.

Exercise 13.11 HARD

Find the flaw: “Theorem: $a^n = 1$ for every $a \neq 0$ and $n \geq 0$. Base: $a^0 = 1$. Strong step: $a^{n+1} = \frac{a^n \cdot a^n}{a^{n-1}} = \frac{1 \cdot 1}{1} = 1$.”

Solutions to the Exercises

Solution 13.1. Base: $1 = 1^2$. Step: $\sum_{i=1}^{n+1} (2i - 1) = n^2 + (2n + 1) = (n + 1)^2$. ■

Solution 13.2. Base: $1 \cdot 1! = 1 = 2! - 1$. Step: $\sum_{k=1}^{n+1} k k! = (n + 1)! - 1 + (n + 1)(n + 1)! = (n + 1)!(n + 2) - 1 = (n + 2)! - 1$. ■

Solution 13.3. Base: $\frac{1}{2} = \frac{1}{2}$. Step: $\frac{n}{n+1} + \frac{1}{(n+1)(n+2)} = \frac{n(n+2)+1}{(n+1)(n+2)} = \frac{(n+1)^2}{(n+1)(n+2)} = \frac{n+1}{n+2}$. ■

Solution 13.4. Base: $7! = 5040 > 2187 = 3^7$. Step: for $n \geq 7$, $(n + 1)! = (n + 1)n! > (n + 1)3^n \geq 8 \cdot 3^n > 3^{n+1}$. ■

Solution 13.5. Base: $6 \mid 0$. Step: $(n + 1)^3 - (n + 1) = (n^3 - n) + 3(n^2 + n) = (n^3 - n) + 3n(n + 1)$. The first term is divisible by 6 by hypothesis; $n(n + 1)$ is even, so $3n(n + 1)$ is divisible by 6 as well. ■

Solution 13.6. Base: $1 = 1$. Step: with $T_n = \frac{n(n+1)}{2}$, $\sum_{k \leq n+1} k^3 = T_n^2 + (n + 1)^3 = \frac{(n+1)^2}{4}(n^2 + 4(n + 1)) = \frac{(n+1)^2(n+2)^2}{4} = T_{n+1}^2$. ■

Solution 13.7. Base: $(1 + x)^0 = 1 \geq 1$. Step: $(1 + x)^{n+1} = (1 + x)^n(1 + x) \geq (1 + nx)(1 + x)$ — this multiplication preserves the inequality *because* $1 + x \geq 0$, i.e. $x \geq -1$. Expanding: $1 + (n + 1)x + nx^2 \geq 1 + (n + 1)x$. ■

Solution 13.8. $P(s)$: any bar of s squares needs $s - 1$ breaks. Base $s = 1$: 0 breaks. Step: a bar of $s + 1$ squares is first broken into pieces of a and b squares, $a + b = s + 1$, $a, b \leq s$. By strong hypothesis they need $a - 1$ and $b - 1$ further breaks: total $1 + (a - 1) + (b - 1) = s = (s + 1) - 1$, independent of the chosen break. ■

Solution 13.9. Base: $1 = 2^0$. Step: let $n + 1 \geq 2$. If $n + 1$ is even, write $n + 1 = 2m$ with $1 \leq m \leq n$; by strong hypothesis $m = \sum b_i 2^i$, so $n + 1 = \sum b_i 2^{i+1}$. If odd, $n + 1 = 2m + 1$ with $0 \leq m \leq n$, and $n + 1 = 1 + \sum b_i 2^{i+1}$, the new bit $b_0 = 1$. ■

Solution 13.10. Let $S \subseteq \mathbb{N}$ have no least element; set $P(n)$: “no element of S is $\leq n$ ”. $P(0)$: if $0 \in S$ it would be least, so $0 \notin S$. Step: assume $P(n)$, i.e. $S \cap \{0, \dots, n\} = \emptyset$. If $n + 1 \in S$ it would be the least element — impossible; so $P(n + 1)$. By induction every n avoids S : $S = \emptyset$. Contrapositive: every nonempty S has a least element. ■

Solution 13.11. The step at $n = 0$ computes $a^1 = \frac{a^0 a^0}{a^{-1}}$, invoking the hypothesis at $n - 1 = -1$, which is outside the induction’s range (and the base only covers $n = 0$). The chain $P(0) \Rightarrow P(1)$ is broken, so nothing beyond the base is established. ■